

Memory as a Learned Policy:

Co-Training Read and Write Operations for Long-Horizon LLM Agents

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Abstract

Long-horizon LLM agents need a memory store. Today, every system that ships one—Mem0, Letta, Mastra, Zep, our own PIE—hand-codes the rules for what to write and how to read. They use thresholds, prompts, and heuristics. None of those rules learn from outcomes.

We turn memory operations into the actions of a small learned policy. The policy is a LoRA adapter on a 4B parameter language model. We train two of them: a write policy that decides what to store from each turn of conversation, and a read policy that decides how to search memory for an answer. They share the same op vocabulary. They are trained jointly with GRPO [16]. The reward for the read policy is whether the answer is correct. The reward for the write policy is whether the read policy can later answer questions that depend on the turn the write policy saw.

We show that this works on LoCoMo [8] and LongMemEval [19]. **[TBD: report final numbers once Phase A and Phase B run to completion.]** The trained read policy outperforms a heuristic teacher by **[TBD: X]** points. Joint training adds **[TBD: Y]** more points. The same policy works across three different memory backends—a flat chunk store, an entity-based knowledge graph (PIE), and Mastra-style observational memory—with **[TBD: Z]**% drop on transfer.

The contribution is not a new memory store. It is a learned policy that uses the store. Facts stay in the store, where they belong. The policy encodes the strategy, which transfers between users.

1 Introduction

Long-running AI assistants need long-running memory. ChatGPT can remember your preferences. Claude can keep notes across a project. A coding agent should know what design choices it made yesterday. A customer-support bot should know what the customer is angry about now and what they were angry about three weeks ago.

Each of these uses a memory system. Each system has a way to write things down and a way to read them back. And each was hand-built.

The hand-coded gap. Look at the popular open-source memory systems for LLM agents:

- **Mem0** [3] stores facts as flat key-value pairs. When a new fact arrives, an LLM-as-judge prompt picks one of ADD/UPDATE/DELETE/NONE. The thresholds are fixed.
- **Letta** / **MemGPT** [12] treats memory as operating-system tiers. The LLM uses fixed function calls to move things between tiers. The promotion rules are hard-coded.
- **Zep** / **Graphiti** [14] keeps a temporal knowledge graph. Edges have valid times. The invalidation rules are fixed.

- **Mastra** [9] uses two background agents, Observer and Reflector, to compress the chat into a dense observation log. The compression schedule is fixed.
- **Our own PIE** keeps typed entities with state transitions. A 3-tier resolver decides if a new mention matches an existing entity. The thresholds are fixed: 0.85 for string match, 0.85 for embedding accept, 0.65 for reject.

In every case, the rules for what to write and how to read are written by humans. None of the rules update when the system makes a mistake. None of them improve from being used.

What we do. We treat memory operations as actions of a small learned policy. The policy is a low-rank adapter (LoRA) [5] on top of a 4B-parameter open-source language model. The actions are tool calls into a memory backend.

We define one action vocabulary. It has read ops (`memory_search`, `memory_expand`, `memory_filter`, `memory_rerank`) and write ops (`lookup_entity`, `create_entity`, `update_state`, `merge_entities`, `add_relation`, `mark_contradiction`, `forget`, `noop`, `add_alias`). We train two policies.

The *read* policy answers questions. Its reward is whether the answer is correct, minus a small cost for each tool call. We train it with GRPO [16], the simplification of PPO that powers DeepSeek-R1.

The *write* policy ingests conversations turn-by-turn. Its reward is *deferred*: after a write trajectory finishes, we run the read policy on a battery of held-out questions whose answer depends on the turn the write policy just saw. The reward is the read policy’s accuracy on that battery, minus a storage cost.

The two policies are trained jointly. We alternate. Train the read policy against memory the write policy made. Then freeze the read policy and train the write policy against the read policy’s eval. Repeat.

This is the AlphaZero idea applied to memory. Read and write co-adapt. The storage substrate stays swappable.

Why this matters. For a paper, the contribution is the *learned operating system* for memory: facts in the store, strategy in the weights. For a product, it means one trained adapter can serve many users. The user’s facts live in their personal store. The adapter encodes how to use any memory store, no matter whose. That is what generalises. That is what transfers.

Contributions.

1. A backend-agnostic vocabulary of memory operations that compiles to three different storage representations: a flat chunk store, a knowledge graph, and an observational text log.
2. A co-training algorithm that alternates GRPO updates on read and write policies, with the read policy serving as the deferred-reward judge for the write policy.
3. A reward design for the write policy that uses the existing LoCoMo evidence labels as a per-turn counterfactual signal, with no extra annotation needed.
4. Empirical results on LoCoMo and LongMemEval showing [TBD: summarise once experiments land].

2 Related Work

2.1 Memory systems for LLM agents

The space splits along two axes: where facts live (in the model weights or in an external store), and how the operations on those facts are controlled (hand-coded or learned).

Facts in an external store, hand-coded ops. Most production systems sit here. **Mem0** [3] stores facts in a vector index and uses an LLM-as-judge prompt for each new turn to choose between ADD, UPDATE, DELETE, or NONE. **Letta** (formerly MemGPT) [12] keeps a tiered memory (working, recall, archival) and uses LLM function calls to move things between tiers. **Zep** / **Graphiti** [14] stores a bi-temporal knowledge graph—each edge has a valid-time interval and an ingestion-time. **Mastra Observational Memory** [9] uses two background agents (Observer and Reflector) to compress the chat history into a dense bullet-point observation log; it reports the highest published score on LongMemEval (94.87% with gpt-5-mini). **Generative Agents** [13] run periodic reflection cycles that condense raw observations into higher-level beliefs. In all of these, the schedule, thresholds, and prompts that decide what to remember are designed by hand.

Facts in the weights. A different line modifies model parameters per user or per fact. **ROME** / **MEMIT** [10] do surgical edits to feed-forward layer activations to insert specific facts. Per-user fine-tuning embeds a user’s history into a LoRA adapter. These approaches give very tight retrieval (no search needed—the model just knows) but fragment the model fleet (one adapter per user) and risk catastrophic forgetting.

Facts in the store, ops learned. This cell is the smallest and is where we sit. The closest precedent is **ACE** [1] (under review at ICLR 2026), which evolves text-form “playbooks” based on agent execution feedback—but the playbook is a single prompt document, not a structured memory store. **Voyager** [18] maintains a Minecraft skill library that grows over time, with the LLM deciding what to add—but additions are trial-and-error, not RL-trained.

2.2 Reinforcement learning for tool use

The general framework is: an LLM emits tool calls as actions, an environment runs them and returns observations, the episode ends with a verifiable reward, and we train the LLM to emit better tool sequences.

RLHF [11] pioneered using RL on language model outputs, but with reward models trained from human preferences and PPO as the algorithm. **PPO** [15] requires a critic network to estimate baseline rewards.

GRPO [16] drops the critic. Instead, for each prompt, sample a group of G rollouts, compute the reward of each, and use the group mean as the baseline. The advantage of rollout i is $(r_i - \bar{r})/\text{std}(r)$. This works exceptionally well when the reward is verifiable (e.g. correct answer to a math problem). DeepSeek-R1 demonstrated GRPO on chain-of-thought; it is now the default RL algorithm for tool-use training.

Search-R1 [6] applies GRPO to multi-turn search-then-answer over Wikipedia. Each rollout is a sequence of search queries and a final answer. Reported gains: +41% over RAG baselines on NaturalQuestions with Qwen2.5-7B. The Tinker reproduction we built on exceeds the original paper’s numbers across all four QA benchmarks. Our work extends this to the *memory* setting and adds a write side.

Reflexion [17] and verbal RL approaches update natural-language “self-reflections” rather than weights. They are fast and cheap but plateau quickly because the policy is a static prompt.

Curriculum learning [2] is a classic finding we make use of: starting an RL agent on easy tasks gives reward signal that bootstraps learning on harder tasks. We use LoCoMo’s question categories as a curriculum (easy single-hop $\rightarrow$ hard multi-hop and adversarial).

2.3 Joint training of two policies

CoSearch [1] (cited as concurrent ICLR work) jointly trains a reasoning policy and a document ranker via RL on agentic search. **HippoRAG** [4] uses Personalised PageRank over a graph of memories—inspired by the hippocampus—but uses no RL. Our approach extends the joint-training idea from query-time (reasoning+ranking) to ingestion-time (write+read).

2.4 Adapter-based fine-tuning

LoRA [5] adds small low-rank matrices alongside the attention layers of a frozen base model. Fine-tuning only the adapter keeps the base intact and produces a ~ 50 –200 MB artefact that can be applied to any copy of the same base. We use rank-32 adapters on Qwen3-4B-Instruct [7], trained via Tinker—Thinking Machines Lab’s managed RL training service that exposes the right primitives (forward-backward, sample, optim-step) for GRPO.

2.5 What is missing

Despite a crowded space, no published system combines all three:

1. Discrete, interpretable memory operations (Mem0/Letta/PIE have these).
2. Operations chosen by a learned policy supervised by outcome reward (Search-R1 has this for retrieval only).
3. A write side trained against deferred QA accuracy on a held-out battery (no published system).

That is the gap we close.

3 Method

3.1 The op vocabulary

We define a unified action space $\mathcal{A} = \mathcal{A}_R \cup \mathcal{A}_W$. Each action is a JSON tool call with a fixed schema.

Read ops $\mathcal{A}_R$.

- `memory_search(query, k, source)`: Retrieve top- k chunks by lexical (BM25), semantic (dense), or hybrid (RRF [16]) match. The retrieval substrate is whichever Backend the env was built with.
- `memory_expand(seed_uids, k_per)`: Pull adjacent chunks of seeds (one-hop graph follow / sibling-turn lookup).
- `memory_filter(predicate, value)`: Restrict the current hit set by session, speaker, or time window.

- `memory_rerank(strategy, query)`: Re-order hits by dense similarity, recency, or session order.

The episode ends when the model emits a final answer formatted as “**Answer:** ...”.

Write ops $\mathcal{A}_W$.

- `noop`: This turn is not memory-worthy. The default.
- `lookup_entity(query, type, top_k)`: Find existing entities by name. The policy uses this to avoid duplicates.
- `lookup_relation(source_uid, target_uid)`: Inspect existing graph edges.
- `create_entity(name, type, state)`: Add a new entity. Type is one of nine: `person`, `project`, `tool`, `organization`, `belief`, `decision`, `concept`, `period`, `event`, `goal`.
- `update_state(uid, new_state, transition_type, trigger_summary)`: Mutate an entity. Transition type is one of `update`, `contradiction`, `resolution`, `archival`.
- `merge_entities(canonical_uid, alias_uid)`: Collapse two entities, moving transitions and edges.
- `add_relation(source_uid, target_uid, rel_type, description)`: Add a typed graph edge.
- `mark_contradiction(uid, contradicting_state)`: Flag that the current turn conflicts with the entity’s prior state, but do not pick a winner.
- `forget(uid, reason)`: Soft-delete (archive). Keeps the transition history but marks the entity inactive.

3.2 Backend abstraction

The op vocabulary is the invariant. Each backend is an adapter that compiles ops to native operations:

- **FlatBackend**: an in-memory list of windowed text chunks with BM25 index and dense embeddings. `memory_search` is BM25/dense/RRF [16]. Most write ops are no-ops; `create_entity` appends a chunk.
- **PIEBackend**: wraps our PIE `WorldModel`—typed entities, typed transitions, and a relationship graph. Every write op compiles to a real KG mutation.
- **MastraBackend**: an Observer/Reflector pipeline that compresses raw turns into a dated bullet observation log [9]. Read ops scan the log; write ops mostly no-op (the Observer/Reflector run as background processes).

The same trained policy runs on any of the three. Backend-transfer is one of our main empirical claims.

3.3 Episode definitions

Read episodes. One (question, gold.answer) pair per episode. The model emits read ops and a final answer. Reward:

$$r_R(\tau) = \text{judge}(\text{answer}, \text{gold}) - \lambda_R c_R(\tau)$$

where $c_R(\tau)$ = number of tool calls (each costs 0.005–0.01 reward units) and `judge` returns {0.0, 0.5, 1.0} via gpt-4o following the LongMemEval paper’s protocol.

Write episodes. One conversation turn per episode (Decision C in our design doc). The model sees the focal turn, two prior turns, and a top- k snapshot of pre-existing entities. It emits up to four write ops. The episode ends. Reward is computed by:

$$r_W(\tau) = \frac{1}{|Q|} \sum_{q \in Q} \text{judge}(R_{\text{frozen}}(q, M_\tau), \text{gold}_q) - \lambda_W c_W(\tau)$$

where Q is a battery of LoCoMo questions whose evidence label includes the focal turn’s dia_id, M_τ is the resulting memory state after applying τ ’s ops, R_{frozen} is the read policy held fixed during this phase, and $c_W(\tau) = \alpha n_{\text{ops}} + \beta n_{\text{lookups}} + \gamma n_{\text{entities}}$.

The choice of Q is critical and uses the existing LoCoMo evidence labels—no new annotation needed. For each turn t , Q is exactly the LoCoMo questions that name t as evidence. This gives a per-turn counterfactual signal: “did W preserve enough information to answer the questions that depend on this turn?”

3.4 Co-training algorithm

INITIALIZE

```
W_0 = LoRA on Qwen3-4B (random or SFT-warmed)
R_0 = LoRA on Qwen3-4B (random or SFT-warmed)
R_frozen for first iteration = HeuristicPolicy
                                (deterministic Python; we have no R_0 yet)
```

```
FOR t in 1..T:                                # T = 5
  # Phase A_t: train R, freeze W
  build memory M with W_{t-1} (or default chunking for t=1)
  for s in 1..S_R:                             # S_R = 200
    sample G read rollouts per (Q, gold)
    GRPO step on R using r_R
  R_t := R after this phase

  # Phase B_t: train W, freeze R
  for s in 1..S_W:                             # S_W = 100
    sample G write rollouts per turn
    for each rollout, run R_t on the held-out Q-battery for cost+reward
    GRPO step on W using r_W
  W_t := W after this phase

  evaluate(R_t, W_t) on held-out splits
  if not_improving: break
```

The cost regulariser keeps the write policy from learning “store everything” as a strategy. The held-out evaluation lets us stop early.

3.5 Implementation

We use Tinker [7], Thinking Machines’ managed RL training service. The base model lives on Tinker’s GPU cluster. We attach a rank-32 LoRA. Forward-backward and Adam-step calls are sent over the API from a CPU-only orchestrator on the user’s laptop.

The memory backend, the env, the tools, and the reward function all run locally on the orchestrator. A typical training step:

1. Tinker samples G rollouts on its cluster.
2. Each rollout’s tool calls execute locally; results are streamed back to Tinker for the next sample.
3. When the rollout terminates, the local reward function runs (judge call to gpt-4o for read; held-out QA battery for write).
4. Datums (model input + sampled tokens + advantages) are sent back to Tinker for forward-backward + optim-step.

The total data volume per step is small: ~ 8 trajectories, each $\sim 1\text{k}$ tokens. Compute and storage stay manageable.

3.6 Engineering notes

Chunking matters more than retrieval. Early experiments used turn-level units (one Backend unit per conversation turn). Multi-hop recall was 0%—the policy could only retrieve isolated turns like “Yes!” or “Bye!”. We switched to overlapping windows of 6 consecutive turns with a date+session header at the top:

```
[1:56 pm on 8 May, 2023 | session 1]
Caroline: Hey Mel!
Melanie: Hey Caroline!
Caroline: I went to a LGBTQ support group yesterday...
```

The window provides enough text for embeddings to be meaningful, the date header lets the answer LLM resolve relative dates (“yesterday”) against an absolute reference, and the overlap (stride 3) prevents important spans from being split across chunk boundaries.

Reward variance. GRPO needs reward variance within a group or it gets no learning signal. With group size $G = 8$ and Tinker’s default sampling temperature (~ 1.0), this is rarely a problem at training scale. We additionally use the cookbook’s `do_group_rollout_and_filter_constant_reward` which drops groups with zero variance entirely. [TBD: Add DAPO-style dynamic resampling: re-sample G more rollouts on zero-variance groups before giving up.]

Curriculum. LoCoMo questions are pre-categorised: single-hop, multi-hop, open-domain, temporal, adversarial. We start the read policy on the union of single-hop and temporal (where the policy can succeed early), and mix in multi-hop and adversarial after the first ~ 50 GRPO steps. The cold-start with hard questions gives mostly zero-reward rollouts and stalls learning.

4 Datasets and Evaluation

4.1 Training data

LoCoMo [8] is the primary training set. It contains 10 long peer-to-peer conversations with ~ 200 questions each (1,986 total). Each question is annotated with category and with an *evidence* list of `dia_ids` whose content the question depends on. The evidence labels are critical—they let us build the per-turn question batteries that the write reward needs.

We split at the conversation level: 6 train, 2 dev, 2 held-out test. Same conversation never appears in two splits.

4.2 Evaluation

We report on three benchmarks:

1. **LoCoMo held-out**: 2 conversations, ~ 400 questions, in-domain (similar distribution to training but unseen conversations).
2. **LongMemEval-S** [19]: 500 questions, $\sim 115k$ tokens / 30–40 sessions per question. Out of distribution—different conversation style, different question types.
3. **TemporalBench** (ours): 6-axis behavioural temporal benchmark testing proactivity, deadline tracking, gap-aware resumption, staleness detection, rhythm recognition, and commitment follow-through. [TBD: Build and release; spec lives at TEMPORAL-BENCH-SPEC.md.]

We use gpt-4o as the judge across all benchmarks, following the LongMemEval paper protocol.

4.3 Baselines

We compare against the strongest publicly-available memory systems:

- **Naive RAG**: dense retrieval over conversation chunks, no learned policy.
- **Mem0** [3].
- **Mastra Observational Memory** [9].
- **Zep** [14].
- **PIE** (our prior system; PIE-temporal).
- **HeuristicPolicy** (our handwritten read-side teacher), bounded by SFT data quality.

5 Experiments

[TBD: Fill in once Phase A and Phase B runs land.]

5.1 Main results

Table 1 reports the headline numbers across LoCoMo and LongMemEval-S.

Table 1: Main results. [TBD: fill]

System	LoCoMo (held-out)	LongMemEval-S	TemporalBench	Cost
Naive RAG	[TBD:]	[TBD:]	[TBD:]	[TBD:]
Mem0	[TBD:]	[TBD:]	[TBD:]	[TBD:]
Mastra OM (gpt-5-mini)	[TBD:]	94.87 (reported)	[TBD:]	[TBD:]
Zep	[TBD:]	[TBD:]	[TBD:]	[TBD:]
PIE (temporal, ours)	[TBD:]	[TBD:]	[TBD:]	[TBD:]
mempol-Phase-A (read trained)	[TBD:]	[TBD:]	[TBD:]	[TBD:]
mempol-Phase-B (W trained, R fixed)	[TBD:]	[TBD:]	[TBD:]	[TBD:]
mempol-Cotrained (R + W joint)	[TBD:]	[TBD:]	[TBD:]	[TBD:]

Note for the reader. The numbers above are pending the runs described in section 3.5. The infrastructure is verified end-to-end (a five-step Phase-A smoke completed successfully on Tinker; the trajectories and metrics are recorded in our run logs). The remaining work is compute, not engineering.

5.2 Per-category breakdown

[TBD: Per-category accuracy on LoCoMo: single-hop, multi-hop, open-domain, temporal, adversarial. Expected pattern: temporal and single-hop are easy for the read-only Phase A; multi-hop and adversarial benefit most from co-training because they need richer write-side organisation.]

5.3 Backend transfer

Train the policy on FlatBackend; evaluate on FlatBackend, MastraBackend, and PIEBackend. [TBD: Fill the 3×3 transfer table.] Hypothesis: within ~ 5 points across backends, demonstrating the policy learned op semantics rather than backend quirks.

5.4 Ablations

We plan five:

- Action-vocabulary size: 4 ops (Mem0-like) vs. 8 vs. 12 (full).
- Cost coefficient λ_W : 0 (no penalty), 0.001, 0.01, 0.1.
- Frozen R for Phase B: HeuristicPolicy vs. Phase-A LoRA vs. larger Phase-A LoRA.
- Curriculum: easy $\rightarrow$ hard vs. random order.
- Group size G : 4 vs. 8 vs. 16.

[TBD: Fill once main results land.]

6 Analysis

6.1 What does the trained read policy actually do?

[TBD: Qualitative trajectory analysis. Pick 5 questions where Phase A beats the heuristic and 5 where it does not. For each, show the heuristic vs. trained trajectory side by side.]

6.2 What does the trained write policy actually do?

[TBD: Show the entity counts and op distribution per category. Hypothesis: the trained W is more aggressive on `create_entity` for proper nouns and more conservative (`noop`) for chitchat, beating the heuristic on both dimensions.]

6.3 Co-training dynamics

[TBD: Plot reward curves across outer iterations $t = 1 \dots T$. Look for the AlphaZero-style monotone improvement signature.]

7 Discussion

7.1 What we learned

The two findings we expect to centre the discussion on:

1. Granularity of the storage substrate matters more than the read policy’s sophistication. Replacing turn-level units with windowed chunks (six turns, stride three) gave us a 0% → [TBD: X]% improvement on multi-hop *before* any RL training.
2. The deferred-reward signal for the write policy is informative but sparse. The cost regulariser is doing as much work as the QA accuracy term in shaping the trained policy.

7.2 Limitations

Single-user training data. The current training set is LoCoMo’s two-speaker peer chats. Real personal-AI deployments involve one user and one assistant with different conversational styles. We do not yet show that policies trained on LoCoMo transfer to those. [TBD: Run on filtered ChatGPT-export data once we have one.]

Mostly synthetic compute. Tinker’s training service is in private beta. Reproducing this paper requires either Tinker access or porting our recipe to verl/OpenRLHF. We provide both code paths.

Cost. A full co-training run is \$1.5–2.5K of Tinker compute. This is high for academic reproducibility. We will release trained checkpoints publicly.

No write-policy curriculum. We curriculumise the read side on LoCoMo’s question categories. The write side has no analogous structure—all turns are equally valid training data. A possible extension is to curriculumise on conversation length or category density.

7.3 Future work

Multi-backend tools at runtime. Each backend exposes only the read ops. A natural extension is to give the policy multiple backends as separate tool sets and let it learn to route: use the KG for entity questions, the chunk store for verbatim quotes, the observation log for distilled summaries. This is the natural sequel.

Longer write episodes. We use per-turn write episodes for clean credit assignment. Per-conversation episodes would let the policy learn write strategies that span turns (e.g., “defer this decision and revisit if it comes up again”). Cost: harder credit assignment and longer episodes.

Other applications. The architecture (op vocabulary, backend abstraction, deferred reward via downstream task accuracy) is not specific to memory. It applies to any tool-use setting where the agent’s actions modify a substrate that future tool-use queries will read. Code search, knowledge-base curation, and agentic workflow authoring all fit this template.

8 Conclusion

We presented a learned memory operating policy for long-horizon LLM agents. Read and write operations are actions of two LoRA adapters trained jointly with GRPO. The reward for the write policy is the read policy’s accuracy on a held-out question battery whose answers depend on the turn the write policy saw—a deferred signal that requires no extra annotation when LoCoMo evidence labels are available.

[TBD: Closing paragraph with the headline number once experiments land.] The key idea: facts in the store, strategy in the weights. The store holds the user’s data. The weights hold the strategy that any user can share.

Reproducibility. All code is at <https://github.com/USERNAME/mempol> (TODO: replace once pushed). The training recipe drops into the `tinker-cookbook` [7] repository as `tinker-cookbook/recipes/memory`

Acknowledgements. [TBD: Add when ready.]

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